

GTEX Sex Differential Expression Analysis in R

Employer-Facing Technical Portfolio Summary

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Metric	Value
Tissues analysed	Aorta artery, breast mammary tissue, brain hippocampus
Samples	432 aorta, 197 brain hippocampus, 459 breast mammary
Gene variables	Approximately 19,000+ per tissue
Aorta artery DEGs	801
Brain hippocampus DEGs	13
Breast mammary DEGs	5,597
Core tools	R, ggplot2, limma, biomaRt, DAVID/KEGG

Project Overview

This project investigated sex-specific differences in gene expression across clinically relevant human tissues using GTEx v8 transcriptomic data. The employer-facing portfolio version focuses on transferable data analyst capability: data validation, high-dimensional modelling, statistical correction, visualisation and evidence-based interpretation.

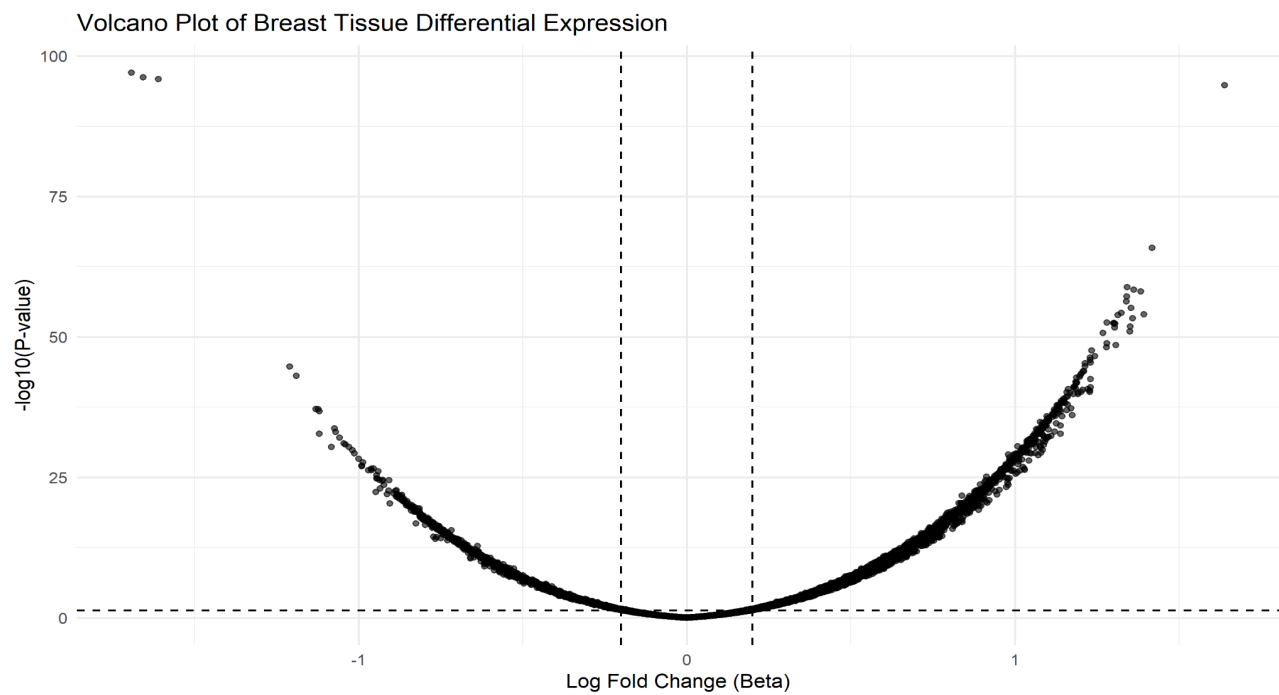
Analytical Workflow

1	Acquire and prepare GTEx expression and phenotype data
2	Validate subject identifiers and merge phenotype metadata with gene expression matrices
3	Model gene expression against biological sex to estimate effect size and significance
4	Apply Benjamini-Hochberg and Bonferroni correction to control false discovery
5	Generate volcano, Manhattan, p-value and gene-type visualisations
6	Interpret KEGG/DAVID pathway enrichment and compare with GWAS disease-linked genes

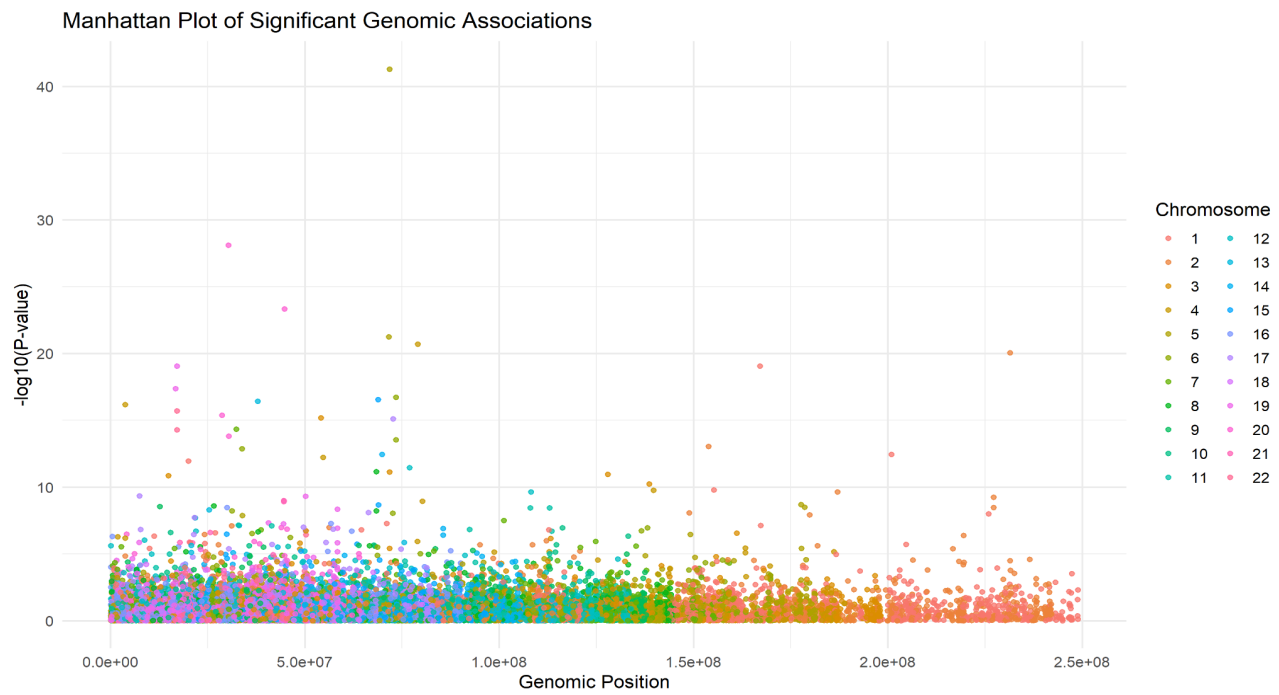
Key Findings

The aorta artery and breast mammary tissues showed stronger evidence of sex-biased expression than brain hippocampus. The dissertation reported 801 aorta artery DEGs, 13 brain hippocampus DEGs and 5,597 breast mammary DEGs after the selected correction strategy. Breast tissue pathway analysis identified enriched pathways including ECM-receptor interaction, cell adhesion molecules, viral protein interaction with cytokine and cytokine receptor, coronavirus disease - COVID-19, and basal cell carcinoma.

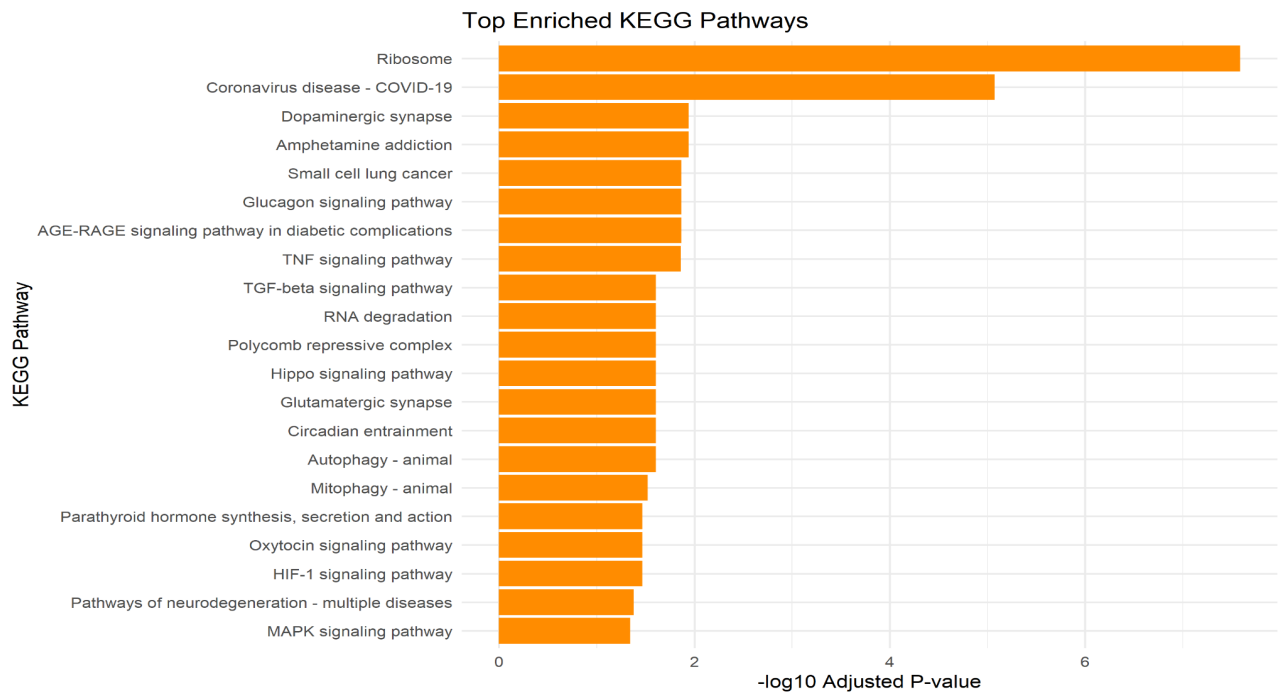
Selected Visual Outputs



Volcano plot showing effect size against statistical significance for breast mammary tissue.



Manhattan plot showing genome-wide distribution of significant expression signals.



Pathway enrichment visualisation summarising biological mechanisms associated with differential expression.

Transferable Employer Value

Although the subject matter is genomic, the project demonstrates transferable skills directly relevant to NHS, public sector and data analyst roles: processing complex structured datasets, validating identifiers, applying formal statistical methods, producing reproducible outputs and presenting complex findings clearly.

Skill Area	Evidence from Project
Data quality	Identifier validation, metadata harmonisation and tissue-level dataset preparation
Statistical analysis	Regression modelling, p-value interpretation and multiple testing correction
Visual analytics	Volcano, Manhattan, p-value and gene distribution visualisations
Insight generation	Pathway enrichment and disease-relevance interpretation
Reproducibility	R scripting, exported outputs and structured project documentation